

Blockchain-based digital traceability for circularity and end-of-life management of PV panels across the value chain, as demonstrated by the SOPHIA project.

Energy × Product traceability and digital product passport × Blockchain and DLT · OS151 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
49 is the world moving	21 can we play, can we win	LOW 1 named in the evidence	€237m partial evidence · per year	LATER research and standards activi...	L3 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity addresses the need for blockchain-based digital traceability to manage the end-of-life of photovoltaic (PV) panels, as demonstrated by the SOPHIA project. It targets energy sector stakeholders—manufacturers, recyclers, and regulators—who must comply with circular economy mandates and ensure transparent, trustworthy data across the value chain. The time is now because regulatory pressure and technological maturity are converging, making digital product passports and blockchain traceability viable and necessary.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€287m €45.3m – €4.4bn	€237m €37.3m – €3.6bn	€296k €47k – €4.6m	partial
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€215k €215k – €575k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Blockchain and DLT	71.5 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

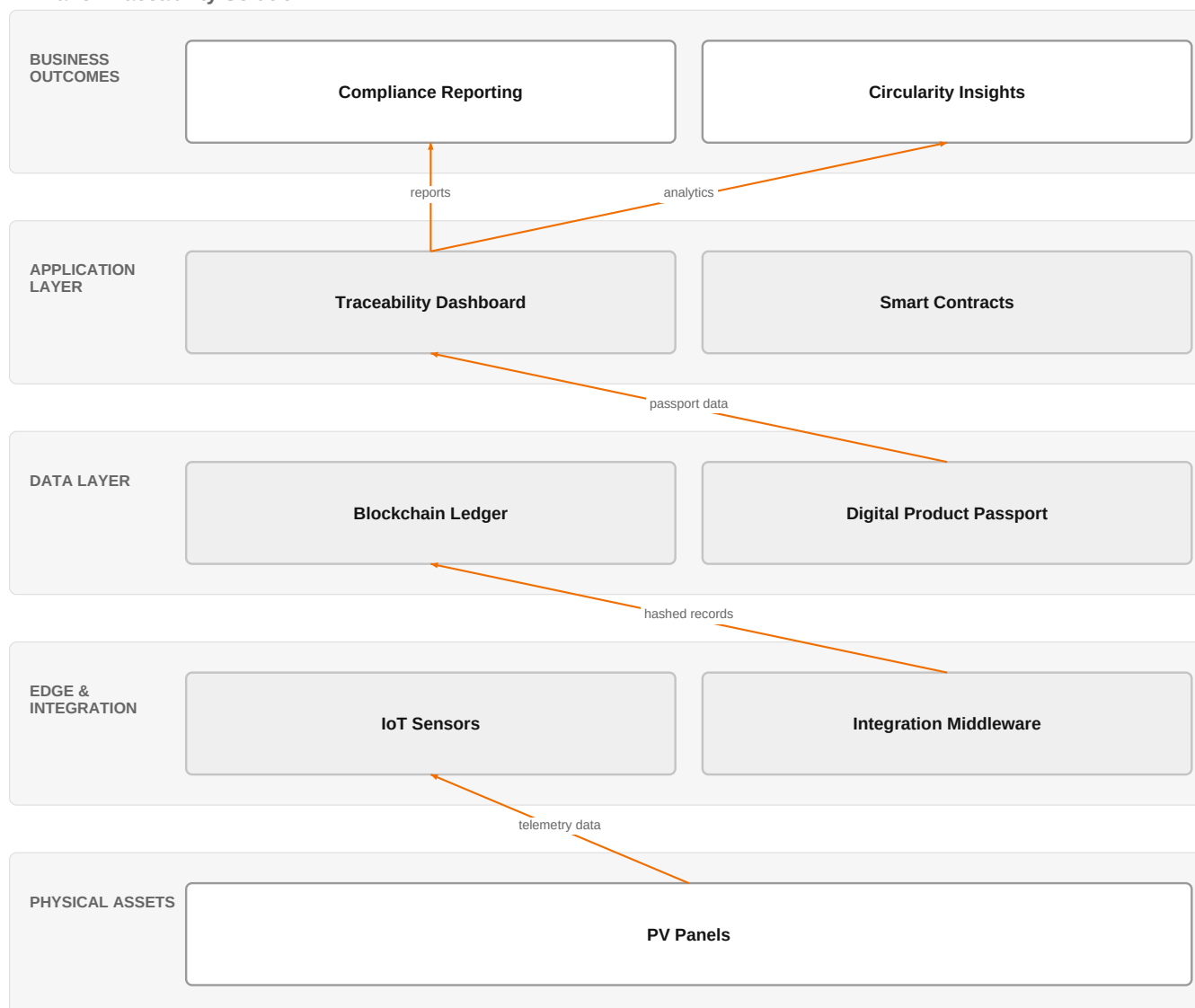
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

PV Panel Traceability Solution



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This solution connects PV panels to a blockchain-backed traceability system, enabling transparent and compliant end-of-life management.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory mandates such as the Ecodesign for Sustainable Product Regulation are pushing for digital product passports to support circular economy goals, particularly for electronics and batteries. Blockchain and IoT integration are maturing, enabling real-time traceability and smart contracts for automated compliance. The SOPHIA project specifically addresses PV panel end-of-life management, highlighting a growing focus on e-waste and circularity in the energy sector.

Sources: SIG-DFE69DA138F9 openalex.org 2026-01-01; SIG-81CEE24B054D openalex.org 2025-12-12; SIG-A2CC51B21A97 cordis.europa.eu 2025-06-01

Who buys, and why now

The primary buyer is a sustainability or compliance officer at a PV panel manufacturer or a large energy utility, who feels the pain of fragmented data and lack of trust across the supply chain. The trigger is an upcoming regulatory deadline or a customer demand for verified circularity claims. They need to prove responsible sourcing and end-of-life handling to avoid

penalties and meet ESG targets.

Why it is hot now — every claim bound to a dated source

- SOPHIA implements advanced digital solutions to increase circularity of PV panels throughout the full value chain, addressing end-of-life management and e-waste awareness. [SIG-A2CC51B21A97]
- The SOPHIA project implements advanced digital solutions to increase circularity of PV panels, addressing end-of-life management and e-waste awareness. [SIG-A2CC51B21A97]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts to collect data from PV panels throughout their lifecycle, and Orange Group researchers to design a blockchain-based traceability framework. The engagement would start with a discovery phase to map the customer's value chain, then deploy IoT sensors and integrate with existing systems, and finally implement a blockchain ledger with smart contracts for automated compliance reporting.

Why Orange can win

Orange can leverage its IoT expertise and research capabilities to deliver a tailored solution, backed by ISO 27001 and SecNumCloud certifications to address sovereignty concerns. However, the assets are thin—there is no specific blockchain platform or energy vertical solution named, so Orange would need to partner or build on third-party technology, which could weaken its position.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Blockchain and DLT	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

LOW competitive intensity (score 2.9; 1 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitors include AWS, which offers blockchain services and has strong cloud infrastructure; Capgemini, a global systems integrator with energy sector expertise; and industrial specialists like Honeywell, Schneider Electric, Siemens, and Nokia, who have deep OT integration capabilities. Telefonica Tech and SpaceX Starlink also have energy offerings. The customer will compare Orange's ability to integrate with existing OT systems and provide end-to-end security.

Sources: SIG-01499FFBAF54 arxiv.org 2026-06-05

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	evidenced	SIG-01499FFBAF54
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—
Telefonica Tech	Telecom operator peer	active in Energy	structural	—
Honeywell	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—

Nokia	Network equipment vendor	active in Energy; competes in OX: Smart Industries	structural	—
SpaceX Starlink	Satellite operator	active in Energy	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently tracking the lifecycle of your PV panels, from manufacturing to end-of-life?
- 2 What specific regulatory or customer requirements are driving your need for traceability?
- 3 Who in your organization is responsible for sustainability reporting and what data do they lack?
- 4 Have you participated in any pilot projects for digital product passports or blockchain traceability?
- 5 What is your current process for managing end-of-life PV panels, and where are the biggest data gaps?

Objections you will meet

They will say	You can answer
We already have a traceability system in place.	That's true, but many existing systems are centralized and lack the trust and transparency that blockchain provides. Our solution can complement your existing systems by adding an immutable layer that satisfies regulatory and customer demands for verifiable data.
Blockchain is overhyped and not proven at scale.	You're right that blockchain has had mixed results, but the technology has matured, and projects like SOPHIA are demonstrating its viability for supply chain traceability. We focus on practical applications that solve real business problems, not just technology for its own sake.
We're concerned about data sovereignty and security.	That's a valid concern. Orange's solutions can be deployed on sovereign infrastructure, and we hold ISO 27001 and SecNumCloud certifications, ensuring your data stays secure and compliant with local regulations.

Risks and unknowns

The main risk is that the SOPHIA project is still in research phase, and there is no proven commercial deployment. Regulatory timelines for digital product passports are uncertain, and the customer may not yet feel urgency. Additionally, the lack of a named blockchain platform in Orange's assets means we may need to rely on partners, which could dilute our value proposition.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: prepare a differentiation narrative for a future tender on PV traceability, anchored in Orange's IoT expertise and blockchain/DLT technology, and outline how ISO 27001 and SecNumCloud certifications would support a secure digital product passport.
Sales	In a conversation with a European solar energy association or a major PV manufacturer's sustainability lead, open with: 'We're tracking the SOPHIA project on digital traceability for PV panels — how is your organisation approaching end-of-life tracking and circularity today?'
Strategist / Innovator	Deep-dive brief: map the SOPHIA project's consortium and technical architecture to identify where Orange's IoT and blockchain capabilities could plug into the PV circularity value chain, and assess the readiness of relevant standards for a later-stage opportunity.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-EC14886FDE80	2026-03-22	openalex.org	1	A Blockchain-Enabled Smart Contract Architecture for Enhancing Transparency, Traceability, and ...
SIG-A85B5F017CD1	2026-02-27	openalex.org	1	Blockchain and Big Data Integration for Transparent and Traceable Supply Chains
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-DFE69DA138F9	2026-01-01	openalex.org	1	Shaping a circular future for electronics: The role of Digital Product Passports
SIG-81CEE24B054D	2025-12-12	openalex.org	1	Blockchain-Driven Supply Chain Automation for Real-Time Traceability
SIG-6786E4AD1713	2025-11-22	openalex.org	1	The quest for more circular battery value chains: Implementing the EU digital battery passport ...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-B429728E0CD0	2025-09-10	openalex.org	1	Blockchain and pricing strategies in supply chains considering consumer traceability preference...
SIG-25647612A208	2025-08-30	openalex.org	1	A Narrative Review and Gap Analysis of Blockchain for Transparency, Traceability, and Trust in ...
SIG-75DE7DFC9EA4	2025-06-01	cordis.europa.eu	1	Empowering Circular Operations in the Automotive Plastics Value Chain
SIG-A2CC51B21A97	2025-06-01	cordis.europa.eu	1	Implementation of Advanced Digital Solutions to increase the circularity of PV panels throughou...
SIG-4674EE3A0873	2025-02-26	The Role of Network Secur...	1	5G-Enabled Smart Healthcare System with the Integration of Blockchain Technology
SIG-01499FFBAF54	2026-06-05	arxiv.org	3	Blockchain Infrastructure for Intelligent Cyber--Physical--Social Systems:Post-Quantum Security...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.